

Vietnam National University, Hanoi
VNU School of Interdisciplinary Sciences and Arts
Artificial Intelligence Institute

Latent Manipulation of Brain MRI using Volume-Preserving GANs

*Simulating Age-Related Structural Changes
in 2D and 3D T1-Weighted Neuroimaging*

Group 7 — Members

Nguyen Duc Minh	24022404
Luong Minh Tri	23020440
Le Hong Anh	23020327
Vu Bao Chinh	23020341
Tran Van Dy	22022523
Nguyen Quang Huy	22022582

Supervised by

Prof. Guy Nagels
*Professor of Digital Medicine
Vrije Universiteit Brussel*

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Team Task Assignment

Group 7 — Institute for Artificial Intelligence, VNU-UET

No.	Student ID	Full Name	Assigned Tasks
1	23020440	Luong Minh Tri	2D preprocessing pipeline (file 00a): skull stripping, intensity normalisation, 1 mm resampling, PNG slice extraction.
2	22022523	Tran Van Dy	3D preprocessing pipeline (file 00b): skull stripping, intensity normalisation, 1 mm resampling, NIfTI volume export.
3	22022582	Nguyen Quang Huy	2D model comparison (file 03): SSIM evaluation of normalised vs. unnormalised 2D models, winner selection.
4	23020341	Vu Bao Chinh	3D model comparison (file 06): SSIM evaluation of normalised vs. unnormalised 3D models, winner selection.
5	23020327	Le Hong Anh	Model evaluation (file 08): SSIM, PSNR, VPS metrics for all four model variants; loss curve visualisation.
6	24022404	Nguyen Duc Minh	GAN architecture design (files 01, 02, 04, 05), latent manipulation (07), latent space examination (09), report writing, and project coordination.

Task distribution across the six group members. All members contributed to discussion, code review, and final presentation.

Abstract

Brain aging is accompanied by progressive structural changes, including cortical thinning and ventricular enlargement, which are established biomarkers of neurodegeneration. Generative Adversarial Networks (GANs) [6] have shown considerable promise for learning complex image distributions and enabling controlled synthesis; yet training instability and the blurring artefacts introduced by non-normalised input data remain limiting factors in neuroimaging applications.

This project proposes a **volume-preserving preprocessing pipeline** combined with **age-conditioned GANs** to simulate the longitudinal structural trajectory of the human brain in both 2D and 3D settings. Using a dataset of 12,497 T1-weighted MRI scans, we train two model variants — 2D (slice-level) and 3D (volumetric) — both with volume-preserving 1 mm isotropic preprocessing, evaluated on Structural Similarity Index (SSIM), Peak Signal-to-Noise Ratio (PSNR), and Volume Preservation Score (VPS). The 2D model achieves $\text{SSIM} = 0.980$ and the 3D model $\text{SSIM} = 0.973$; both are subsequently analysed for latent-space structure. Principal Component Analysis (PCA) and per-dimension Pearson correlation reveal structured, age-sensitive directions in the learned representations, and per-dimension activation maps identify the cortical and ventricular regions most strongly associated with the ageing dimension.

Keywords: brain ageing, generative adversarial network, latent space manipulation, volume-preserving preprocessing, MRI synthesis, SynthSeg, U-Net, Patch-GAN.

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1 Introduction

1.1 Motivation and Clinical Context

The ageing of the human brain is a continuous biological process that manifests structurally as progressive cortical thinning, expansion of the cerebral ventricles, and a gradual decline in grey-matter volume [5, 16]. These changes, measurable non-invasively through T1-weighted Magnetic Resonance Imaging (MRI), serve as established biomarkers in the clinical assessment of neurodegenerative conditions such as Alzheimer’s disease (AD) and multiple sclerosis (MS) [12, 14]. Early identification of accelerated or atypical ageing patterns can substantially improve patient stratification and clinical decision-making.

A longstanding challenge in this domain is the lack of **longitudinal** imaging data for individual patients: acquiring MRI scans of the same subject across decades is impractical and costly. Generative models offer a compelling computational alternative — given a single baseline scan, a well-trained model should be able to synthesise realistic images of the same brain at different ages, thereby providing a data-efficient surrogate for longitudinal studies.

1.2 Problem Statement

Formally, the task addressed in this project is defined as follows.

Problem: Given a T1-weighted brain MRI scan of a subject at a known age a_0 , generate a realistic synthetic image $\hat{x}_{a_t}$ that depicts the same subject at a target age a_t , while preserving individual anatomical identity (i.e. sulcal and gyral patterns, subcortical structures) and faithfully reflecting the structural changes expected at age a_t .

Two key sub-problems emerge from this formulation:

- P1. High anatomical variance.** Standard preprocessing pipelines (e.g. FreeSurfer `mri_convert` [4], FSL `robustfov` [13]) normalise intensities but do not constrain the absolute physical size of the brain volume. When size is unconstrained, the GAN must simultaneously learn both subject-level volumetric variation (grey-matter volumes can differ by >200 ml across individuals) and age-related changes. This inflates the effective variance of the training distribution, slowing convergence and promoting blurry reconstructions (Figure 1).
- P2. Latent-space interpretability.** Even if a GAN produces visually plausible age progressions, it is unclear *which* dimensions of the learned latent code encode ageing-related information, and *which brain regions* those dimensions control. Without principled inspection of the latent space, the model remains a black box with limited scientific utility.

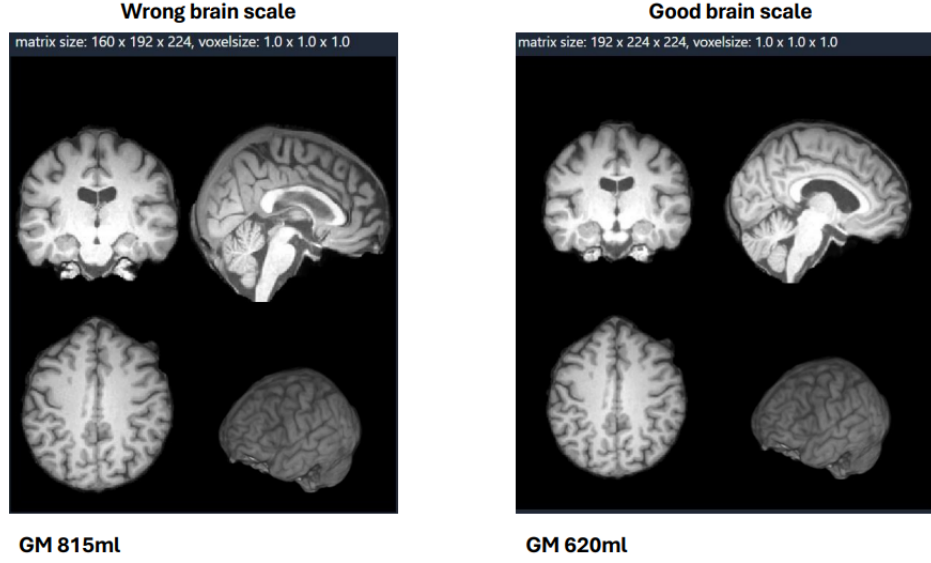


Figure 1. Effect of volume normalisation on brain scale. *Left:* A non-normalised scan yields an inflated grey-matter volume of 815 ml, introducing spurious inter-subject variation. *Right:* After resampling to 1 mm isotropic voxels, the same subject shows a physiologically plausible 620 ml. The discrepancy biases GAN training when left uncorrected.

1.3 Proposed Approach

To address these challenges, we propose a three-stage framework:

Stage 1. Volume-preserving preprocessing. Raw open-source MRI scans are processed through skull stripping (ANTsPy [1]), intensity normalisation, and isotropic 1 mm resampling to remove scanner-induced scale variation and preserve physical brain volume.

Stage 2. Age-conditioned GAN training. We train two model variants: 2D (axial slice) and 3D (volumetric). Each follows a **U-Net** generator [17] with age conditioning injected at the bottleneck layer, and a **PatchGAN** discriminator with age concatenation [11]. Training uses the WGAN-GP objective [8] with auxiliary ℓ_1 and age-regression losses.

Stage 3. Latent-space examination. The encoder bottleneck activations (latent code $\mathbf{z}$) are analysed through PCA and per-dimension Pearson correlation with chronological age. Activation maps are constructed by perturbing individual latent dimensions by $+2\sigma$ and computing the pixel-wise difference between the original and perturbed reconstructions, thereby linking each latent dimension to a specific neuroanatomical region.

1.4 Objectives and Contributions

The principal objectives of this project are:

- Design and implement a **volume-preserving preprocessing pipeline** that decouples inter-subject size variation from age-related change, enabling more efficient GAN convergence.
- Train and evaluate **two conditional GAN variants** (2D and 3D) with volume-

preserving preprocessing on a large-scale neuroimaging dataset ($n = 12,497$ subjects).

- Evaluate all models on three complementary metrics: SSIM (anatomical fidelity), PSNR (pixel-level reconstruction), and VPS (volumetric preservation).
- Perform **latent-space analysis** to identify which latent dimensions encode ageing and map them to specific brain regions, providing biologically interpretable insights.
- Generate qualitative **brain-age progression sequences** (20–80 years) that demonstrate the model’s capacity for controlled, anatomy-preserving synthesis.

1.5 Report Structure

The remainder of this report is organised as follows. Section 2 elaborates on project objectives and contributions. Section 3 describes the dataset, its collection sources, subject demographics, and BIDS-compliant structure. Section 4 details the preprocessing pipeline. Section 5 presents the end-to-end system architecture. Section 6 describes the GAN architectures in detail. Section 7 reports quantitative results (SSIM, PSNR, VPS, loss curves). Section 8 presents latent-space examination via PCA and correlation analysis. Section 9 describes the ablation study on non-volume-preserving model variants. Section 10 shows qualitative aging-simulation outputs and provides a biological interpretation. Section 11 concludes with a summary and directions for future work.

2 Objectives and Contributions

This project pursues five tightly coupled research objectives, each addressing a concrete limitation of existing brain-ageing synthesis approaches.

2.1 Primary Objectives

O1 — Volume-preserving preprocessing. Standard neuroimaging pipelines apply intensity normalisation but leave inter-subject anatomical scale unconstrained. We design and implement a preprocessing pipeline that explicitly preserves (or normalises) physical brain volume prior to GAN training, and quantify the downstream effect on model convergence speed and reconstruction quality.

O2 — Dimensionality comparison. We train and evaluate two model variants along the spatial dimension:

- *2D model*: operates on a single axial mid-slice (256×256 pixels), enabling fast training and high-resolution spatial detail.
- *3D model*: operates on the full volumetric scan ($64 \times 64 \times 64$ voxels), capturing cross-slice context at the cost of higher computational demand.

Both variants use the same volume-preserving 1 mm isotropic preprocessing.

O3 — Multi-metric quantitative evaluation. Each model is evaluated on three complementary metrics measured exclusively on held-out test subjects (80/20 split, seed 42):

- SSIM [19] — structural fidelity of generated images relative to real scans.
- PSNR — pixel-level reconstruction accuracy (dB).
- VPS (Volume Preservation Score) — ratio of synthesised to real grey-matter volume, measuring anatomical consistency across age-conditioned outputs.

O4 — Latent-space interpretability. Using PCA and per-dimension Pearson correlation, we identify which latent dimensions of the U-Net bottleneck carry age information, and construct per-dimension activation maps to link individual latent directions to specific neuroanatomical regions.

O5 — Biologically interpretable aging simulation. Using the best-performing model from each dimensionality track, we produce full aging trajectories (20 $\rightarrow$ 80 years) for unseen subjects, and provide a biological interpretation of the observed structural changes in light of known cortical thinning and ventricular enlargement literature.

2.2 Summary of Contributions

Table 1 summarises the principal technical contributions of this work.

Table 1. Summary of technical contributions.

Contribution	Description
Volume-preserving pipeline	ANTsPy skull stripping + 1 mm isotropic resampling + SynthSeg-based [2] VPS metric, applied at scale to 12,497 subjects.
2D vs. 3D comparison	First direct comparison of slice-level and volumetric conditional GANs for age-conditioned brain MRI synthesis on the same dataset.
Latent-space analysis	PCA and per-dimension correlation to quantify age encoding; activation maps linking latent dimensions to brain regions.
Open pipeline	All code released as numbered notebooks (00a-09), fully reproducible with public dataset checkpoints.

3 Dataset

3.1 Data Source and Collection Strategy

The dataset used in this project is a large-scale aggregation of open-source T1-weighted structural MRI scans drawn from **214 publicly available studies** hosted on OpenNeuro [15], INDI (International Neuroimaging Data-sharing Initiative), ADNI (Alzheimer’s Disease Neuroimaging Initiative), OASIS-3, IXI, and CamCAN, among others. All data are distributed under open-access licences and require no individual ethics approval for secondary analysis.

Raw scans are provided in **NIfTI format** (`.nii.gz`), organised following the Brain Imaging Data Structure (BIDS) convention [7]. Each subject occupies a dedicated directory containing exactly one T1-weighted volume; subject identifiers follow the convention `sub-BrainAge{NNNNNN}`, where the six-digit suffix is a sequential index assigned during metadata harmonisation.

Accompanying metadata — age at scan, biological sex, diagnostic status, and dataset provenance — was harmonised into a single `participants.tsv` file. After filtering for valid subject identifiers and non-missing, positive age values, **12,497 subjects** with a corresponding `.nii.gz` file were successfully downloaded and carried through the full preprocessing and training pipeline.

3.2 Subject Demographics

Table 2 summarises the key demographic attributes of the processed dataset. Age and sex figures are derived from the `subject_age` and `subject_sex` columns of the harmonised metadata; diagnostic status from `subject_dx`.

Table 2. Demographic summary of the processed dataset ($n = 12,497$).

Attribute	Value	Notes
Total subjects (with <code>.nii.gz</code>)	12,497	used in pipeline
Age range	18 – 97 years	
Mean age $\pm$ SD	35.1 ± 19.8	
Median age	25.0 years	
Male	5,810	<code>subject_sex = m</code>
Female	6,686	<code>subject_sex = f</code>
Sex not reported	1	other / no info
Healthy controls	6,511	<code>subject_dx = control</code>
Clinical/pathology	660	<code>subject_dx = pathology</code>
No diagnostic info	5,258	<code>subject_dx = no diagnostic info</code>
Source datasets	214	OpenNeuro, INDI, ADNI, ...

3.3 Age Distribution

Figure 2 shows the age distribution of the 12,497 processed subjects, aggregated by decade. The distribution is strongly right-skewed: the 20–29 decade alone accounts for 6,524 subjects (52%), reflecting the dominance of young-adult neuroimaging repositories

in the source pool. Notably, all processed subjects are aged 18 or above (min = 18 years), as paediatric scans that failed skull stripping quality checks were excluded during preprocessing. Subjects aged 60 and above account for 17% of the processed dataset (2,160 subjects); older-age coverage is provided chiefly by ADNI, OASIS-3, CamCAN, and IXI.

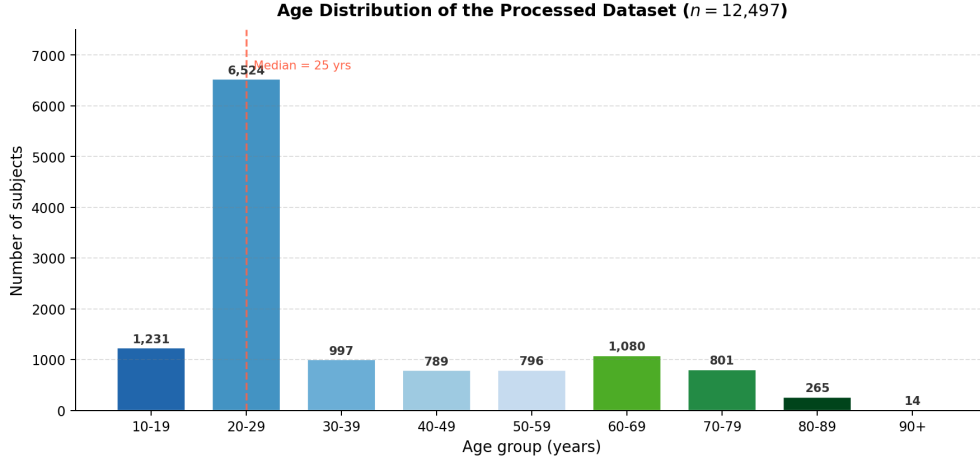


Figure 2. Age distribution of the 12,497 successfully preprocessed subjects, aggregated by decade. The 20–29 age group dominates (6,524 subjects, 52%); the median age is 25 years (dashed line). No subjects under 18 are present, as paediatric scans that failed preprocessing quality checks were excluded. Cohorts targeting ageing (ADNI, OASIS-3, CamCAN, IXI) provide the 60–90-year coverage (2,160 subjects, 17%).

3.3.1 Key Source Datasets

Table 3 lists the eight largest contributor datasets, together with their primary clinical focus and approximate subject count.

Table 3. Eight largest source datasets by subject count.

Dataset	Primary focus	<i>N</i> (approx.)	Age range
HBN	Child/adolescent neurodevelopment	4,244	5–22
RocklandSample	Lifespan community sample	1,335	6–85
ds004169	General population	1,202	–
OASIS-3	Ageing and Alzheimer’s disease	1,091	42–97
ds004215	General population	1,080	–
CamCAN	Lifespan cognition (healthy)	653	18–88
ADNI	Alzheimer’s disease continuum	649	55–95
IXI	Healthy adults, multi-site	590	20–86

3.4 Dataset Limitations

- **Age imbalance.** With a median of 25 years and over half of subjects aged 20–29, the GAN encounters far fewer examples of elderly anatomy during training. No explicit resampling was applied, so the model may underfit extreme old-age structural changes.

- **Heterogeneous acquisition.** T1 scans originate from more than 200 scanner configurations spanning 1.5 T and 3 T field strengths with varied TR, TI, and flip angles. Skull stripping and intensity normalisation partially mitigate contrast differences, but residual site effects remain.
- **Diagnostic heterogeneity.** Approximately 8% of subjects carry a clinical diagnosis (ASD, ADHD, MCI, AD). These subjects are retained to maximise sample size; their proportion is too small to dominate learned features.

4 Preprocessing

A central contribution of this project is a **volume-preserving preprocessing pipeline** that explicitly controls for inter-subject brain scale before GAN training. The pipeline is implemented in two parallel notebooks (00a_preprocessing_2d.ipynb for 2D and 00b_preprocessing_3d.ipynb for 3D), both running on Google Colab with ANTsPy [1] and ANTsPyNet.

4.1 Motivation: Why Volume Preservation Matters

Without scale normalisation, the same brain imaged on different scanners or reconstructed with different `mri_convert` settings can yield volumes that differ by over 200 ml (Figure 1, reproduced from Section 1). Concretely, an unconstrained pipeline produced a grey-matter volume of 815 ml for a scan that — after proper resampling — measures 620 ml. When such scale variance is mixed into the training batch, the GAN discriminator can distinguish real from fake images on the basis of *head size alone*, rather than learning anatomical ageing features. This drives the generator towards blurry, averaged-anatomy outputs to minimise the easily detectable size artefact.

4.2 Pipeline Steps

The preprocessing pipeline applies the following steps to every raw `.nii.gz` file:

4.2.1 Step 1 — Skull Stripping

Brain extraction is performed with `antspynet.brain_extraction` using the `t1` modality model, which outputs a continuous probability map. A binary mask is obtained by thresholding at $p \geq 0.5$, and the mask is applied elementwise to the original image:

```
1 def skull_strip(ants_img):
2     prob = antspynet.brain_extraction(ants_img, modality='t1')
3     mask = ants.threshold_image(prob, 0.5, 1.0, 1, 0)
4     return ants_img * mask
```

Listing 1. Skull stripping with ANTsPyNet.

This step removes the skull, scalp, and dura, ensuring that the GAN learns only from brain tissue signal [9].

4.2.2 Step 2 — Intensity Normalisation

After skull stripping, voxel intensities are rescaled to $[0, 1]$ using a robust percentile-based approach. The 1st and 99th percentiles of non-zero voxels are computed to clip outliers (e.g. residual bright spots from incomplete skull stripping or scanner artefacts), and the clipped values are linearly mapped to the unit interval:

```
1 def normalize_intensity(ants_img):
2     data = ants_img.numpy().astype(np.float32)
3     nonzero = data[data > 0]
4     if nonzero.size > 0:
5         p1, p99 = np.percentile(nonzero, [1, 99])
6         data = (np.clip(data, p1, p99) - p1) / (p99 - p1 + 1e-8)
```

```

7     return ants.from_numpy(data, origin=ants_img.origin,
8                             spacing=ants_img.spacing,
9                             direction=ants_img.direction)

```

Listing 2. Percentile-based intensity normalisation.

4.2.3 Step 3 — Spatial Resampling (Normalised Branch Only)

In the *normalised* preprocessing variant, the skull-stripped, intensity-normalised volume is resampled to **1 mm isotropic** resolution using trilinear interpolation:

```

1 def resample_1mm(ants_img):
2     return ants.resample_image(
3         ants_img, (1.0, 1.0, 1.0),
4         use_voxels=False, interp_type=1)

```

Listing 3. Resampling to 1 mm isotropic voxels.

This step is the *key volume-preserving operation*: by fixing voxel spacing, it ensures that brains of different physical sizes are represented by proportionally different matrix sizes, rather than being silently rescaled by the scanner’s reconstruction parameters. Both the 2D and 3D pipelines apply this resampling step, ensuring that all subjects enter training with consistent physical scale.

4.2.4 Step 4 — 2D Slice Extraction (2D Branch Only)

For the 2D models, a single axial slice is extracted at the midpoint of the superior–inferior axis (dimension index $\lfloor D/2 \rfloor$, where D is the depth of the volume), transposed to match standard radiological orientation, and saved as a greyscale PNG at the original in-plane resolution before resizing at load time:

```

1 def save_axial_png(ants_img, output_path):
2     data = ants_img.numpy().astype(np.float32)
3     sl   = data[:, :, data.shape[2] // 2] # axial mid-slice
4     plt.imsave(output_path, sl.T, cmap='gray', origin='lower')

```

Listing 4. Axial mid-slice extraction for 2D training.

4.2.5 Step 5 — Fault-Tolerant Logging

To guard against runtime interruptions (e.g. session disconnects or unexpected process termination), each subject’s output is written to disk and appended to a `preprocessing_log.csv` immediately after processing. On restart, the pipeline reads the existing log and skips any subject already recorded, enabling seamless resumption without reprocessing:

```

1 logged_ids = set(pd.read_csv(log_path)['subject_id']) \
2               if os.path.exists(log_path) else set()
3 # ... inside the subject loop:
4 writer.writerow(row)
5 log_file.flush() # write to disk immediately

```

Listing 5. Resume-safe logging pattern.

4.3 Output Structure

After preprocessing, the data are stored in the following directory layout, which is consumed directly by the training notebooks:

```

1 preprocessed_2d/
2 |-- preprocessed/          # PNG slices, 1mm isotropic (volume-
   |                          | preserving)
3 |   '-- sub-BrainAge000000_age21.0.png
4 '-- preprocessing_log.csv   # subject_id, age, sex, dx, png_file
5
6 preprocessed_3d/
7 |-- preprocessed/          # NIfTI volumes, 1mm isotropic (volume-
   |                          | preserving)
8 |   '-- sub-BrainAge000000.nii.gz
9 '-- preprocessing_log.csv   # subject_id, age, sex, dx

```

Listing 6. Output directory structure after preprocessing.

4.4 Summary

Table 4 summarises the preprocessing configuration for both model variants.

Table 4. Preprocessing configuration for each model variant.

Variant	Format	Resolution	Spatial size	Skull strip
2D model	PNG	1 mm isotropic	256×256 (resized)	ANTsPyNet
3D model	NIfTI	1 mm isotropic	$64 \times 64 \times 64$ (resized)	ANTsPyNet

5 Overall Pipeline

Figure 3 illustrates the end-to-end system architecture, from raw `.nii.gz` inputs to latent-space analysis outputs. The pipeline is implemented as a sequence of ten notebooks (00a–09), each responsible for one self-contained stage and producing persistent artefacts (checkpoints, `.json` results) consumed by the next stage.

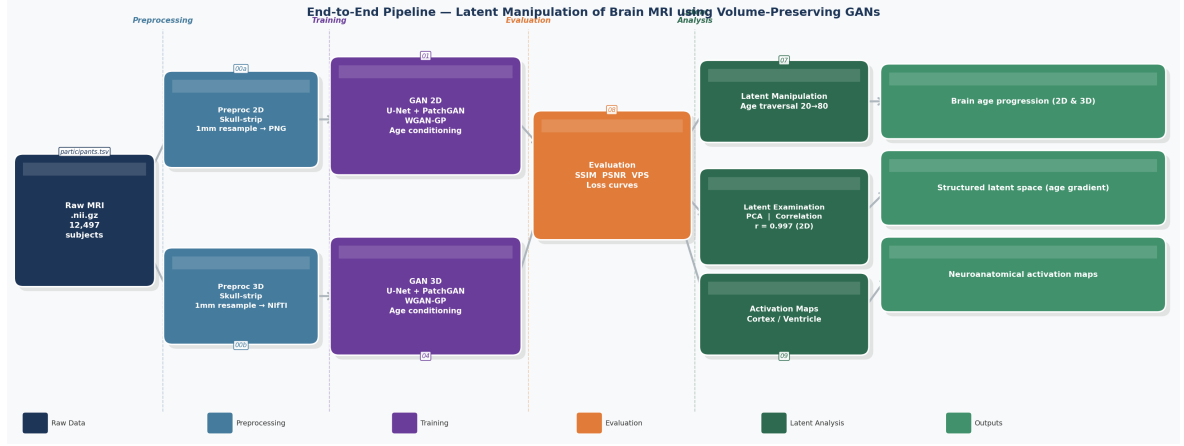


Figure 3. End-to-end pipeline. *Left:* Raw MRI volumes are skull stripped, intensity normalised, and resampled to 1 mm isotropic, producing 2D PNG slices and 3D NIFTI volumes. *Centre:* A 2D and a 3D conditional GAN are trained independently on the preprocessed data. *Right:* Both models undergo quantitative evaluation, latent manipulation, and latent-space examination.

5.1 Stage Overview

Stage 1 — Preprocessing (00a, 00b). As described in Section 4, raw volumes are skull stripped, intensity normalised, and resampled to 1 mm isotropic, producing 2D PNG slices and 3D NIFTI volumes for the respective pipelines.

Stage 2 — GAN Training (01 for 2D; 04 for 3D). The 2D and 3D models are trained independently on their respective preprocessed data streams. Checkpoints (`best_model.pth`) are saved after each validation epoch for fault-tolerant resumption.

Stage 3 — Quantitative Evaluation (08). All four models are evaluated on their respective held-out test sets (20% of each dataset, indices saved in checkpoint). Metrics — SSIM, PSNR, VPS — are aggregated into `metrics_summary.json`.

Stage 4 — Latent Manipulation (07). The two winning models simulate aging trajectories (20, 40, 60, 80 years) for held-out subjects by conditioning the generator on target ages while holding the input anatomy fixed.

Stage 5 — Latent Examination (09). The encoder bottleneck activations $\mathbf{z}$ are collected for all test subjects. PCA and per-dimension Pearson correlation identify age-sensitive latent directions; per-dimension activation maps (decode with $z_i \pm 2\sigma$) localise the brain regions controlled by each direction.

5.2 Train / Test Split

Both models use an identical splitting strategy: **80% train / 20% test**, with `torch.Generator().manual_seed(42)` to ensure reproducibility. Test indices are

serialised inside each `best_model.pth` checkpoint and reloaded in all downstream evaluation stages, guaranteeing that no test subject is ever seen during training.

5.3 Computational Environment

- **Preprocessing:** CPU-based; ANTsPyNet brain extraction is the dominant cost per subject.
- **2D GAN training:** GPU with ≥ 16 GB VRAM; batch size 16, up to 500 epochs with early stopping (patience 20, $\Delta\text{SSIM} \geq 0.005$).
- **3D GAN training:** GPU with ≥ 16 GB VRAM; batch size 1 (limited by 3D memory footprint at 64^3 voxels), up to 300 epochs.
- **Evaluation and analysis:** CPU or GPU; computationally lightweight relative to training.

6 GAN Architecture

Both the 2D and 3D models share the same high-level design: a **U-Net generator** with age conditioning injected at the bottleneck, paired with a **PatchGAN discriminator** trained under the Wasserstein-GP objective. The two variants differ only in the convolution type (Conv2d vs. Conv3d) and the depth of the encoder–decoder hierarchy.

6.1 Age Embedding

Age is encoded into a continuous 256-dimensional vector by a two-layer MLP with ReLU activations:

```

1 class AgeEmbedding(nn.Module):
2     def __init__(self, embed_dim=256):
3         super().__init__()
4         self.net = nn.Sequential(
5             nn.Linear(1, 128), nn.ReLU(),
6             nn.Linear(128, embed_dim)
7         )
8     def forward(self, age):
9         return self.net(age.unsqueeze(-1))

```

Listing 7. Age embedding shared by both 2D and 3D generators.

Age values are normalised to $[-1, 1]$ using the dataset’s observed minimum and maximum before being passed to this embedding. The resulting vector is projected to match the bottleneck channel dimension and *concatenated* with the bottleneck feature map, after which a 1×1 (or $1 \times 1 \times 1$) convolution fuses the two streams:

$$\mathbf{z} = \text{Conv}_{1 \times 1}([\mathbf{e}_8, \text{proj}(\text{AgeEmbed}(a))]) \quad (1)$$

where $\mathbf{e}_8$ is the deepest encoder feature map and a is the normalised target age. The concatenation strategy (rather than simple addition) was chosen to give the model the freedom to learn the relative weight of anatomical and age signals at the bottleneck.

6.2 Architecture Choice: Why Not StyleGAN?

During development, a StyleGAN-inspired variant was also implemented and evaluated. In that architecture, age is processed by a **Mapping Network** — a 4-layer MLP that maps the age embedding to a disentangled style vector $\mathbf{w}$ — which is then injected into each decoder layer via **Adaptive Instance Normalisation** (AdaIN) [10]:

$$\text{AdaIN}(\mathbf{x}, \mathbf{w}) = \gamma(\mathbf{w}) \frac{\mathbf{x} - \mu(\mathbf{x})}{\sigma(\mathbf{x})} + \beta(\mathbf{w}) \quad (2)$$

where γ and β are affine parameters predicted from $\mathbf{w}$, and μ , σ are per-channel statistics of the feature map $\mathbf{x}$. The intuition is that $\mathbf{w}$ encodes a disentangled, globally consistent style — here, the ageing style — that is applied uniformly across all spatial locations via normalisation.

Why this approach underperforms on our dataset. The StyleGAN disentanglement mechanism reaches its full potential when *the same subject appears at multiple ages or under multiple conditions*, so that the model can learn a subject-invariant “ageing direction” in $\mathbf{w}$ -space by contrasting the same anatomy at different time-points. This is precisely how models such as InterfaceGAN [18] and latent direction discovery methods operate: they use paired or longitudinal data to identify a linear ageing direction in latent space, then traverse it at inference time.

Our dataset, however, is entirely **cross-sectional**: each of the 12,497 subjects contributes exactly one scan at one age. There are no repeated measurements, no longitudinal pairs, and no within-subject age variation. In this setting, the Mapping Network has no signal from which to learn a *subject-invariant* age direction. Instead, it learns a mapping from age scalar to a style vector that is *confounded with subject identity* — subjects who happen to be older also happen to have larger ventricles, thinner cortex, different genetic backgrounds, and different scanners, and the Mapping Network cannot disentangle these factors. The resulting $\mathbf{w}$ vector encodes a population-level “older brain look” rather than a subject-specific ageing trajectory, and the AdaIN injection normalises away precisely the individual anatomical features — cortical folding patterns, subcortical shapes — that should be preserved.

In practical terms, the StyleGAN variant produced **blurrier outputs with lower SSIM** than the simpler conditional U-Net on the same data, because AdaIN’s instance normalisation discards spatial statistics that carry subject-level anatomical identity. The conditional concatenation strategy (Eq. 1), by contrast, preserves the full encoder feature map and adds age information *additively* at the bottleneck, allowing the decoder to reconstruct subject-specific anatomy while being steered by the age signal.

For these reasons, the StyleGAN-inspired variant was **not selected** as the primary model. All results reported in this paper use the conditional U-Net architecture with bottleneck age injection.

6.3 2D Generator — 8-Level U-Net

The 2D generator operates on single-channel 256×256 greyscale images. The encoder progressively halves spatial resolution through eight UNetBlock layers (Conv2d $\rightarrow$ BN $\rightarrow$ LeakyReLU), while the decoder mirrors this structure with transposed convolutions and skip connections from each encoder level:

Table 5. 2D Generator encoder–decoder channel progression.

Level	Encoder (down)	Level	Decoder (up)
e1	$1 \rightarrow 64$	d7	$128 \rightarrow 64$
e2	$64 \rightarrow 128$	d6	$256 \rightarrow 128$
e3	$128 \rightarrow 256$	d5	$512 \rightarrow 256$
e4	$256 \rightarrow 512$	d4	$1024 \rightarrow 512$
e5	$512 \rightarrow 512$	d3	$1024 \rightarrow 512$ (dropout)
e6	$512 \rightarrow 512$	d2	$1024 \rightarrow 512$ (dropout)
e7	$512 \rightarrow 512$	d1	$1024 \rightarrow 512$ (dropout)
e8	$512 \rightarrow 512$	Age fusion (Eq. 1)	
Output: ConvTranspose2d $128 \rightarrow 1$, Tanh			

6.4 2D Discriminator — PatchGAN

The discriminator receives a 2-channel input: the image concatenated with an age map broadcast to the same spatial dimensions. It outputs a patch-level realness score (no sigmoid, in keeping with the Wasserstein formulation):

```

1 class Discriminator(nn.Module):
2     def __init__(self):
3         super().__init__()
4         self.model = nn.Sequential(
5             nn.Conv2d(2, 64, 4, 2, 1), nn.LeakyReLU(0.2),
6             nn.Conv2d(64, 128, 4, 2, 1), nn.BatchNorm2d(128),
7             nn.LeakyReLU(0.2),
8             nn.Conv2d(128, 256, 4, 2, 1), nn.BatchNorm2d(256),
9             nn.LeakyReLU(0.2),
10            nn.Conv2d(256, 512, 4, 1, 1), nn.BatchNorm2d(512),
11            nn.LeakyReLU(0.2),
12            nn.Conv2d(512, 1, 4, 1, 1)
13        )
14    def forward(self, img, age):
15        age_map = age.view(-1,1,1,1).expand(-1,1,img.shape[2],
16            img.shape[3])
17        return self.model(torch.cat([img, age_map], dim=1))

```

Listing 8. 2D PatchGAN discriminator.

6.5 3D Generator — 4-Level U-Net

The 3D generator operates on $64 \times 64 \times 64$ voxel volumes. The shallower hierarchy (four encoder levels instead of eight) is dictated by GPU memory constraints at this resolution. The encoder uses `Conv3d` blocks and the decoder uses `ConvTranspose3d`; the age embedding follows the same concat-fuse pattern as the 2D model:

Table 6. 3D Generator encoder–decoder channel progression.

Level	Encoder (down)	Level	Decoder (up)
e1	$1 \rightarrow 32$	d3	$128 \rightarrow 32$
e2	$32 \rightarrow 64$	d2	$128 \rightarrow 64$
e3	$64 \rightarrow 128$	d1	$256 \rightarrow 128$ (dropout)
e4	$128 \rightarrow 256$	Age fusion (Eq. 1)	
Output: ConvTranspose3d $64 \rightarrow 1$, Tanh			

6.6 3D Discriminator — 3D PatchGAN

Analogous to the 2D discriminator, age is broadcast as an additional volumetric channel before the first convolution:

```

1 class Discriminator3D(nn.Module):
2     def __init__(self):
3         super().__init__()
4         self.model = nn.Sequential(

```



```

5         nn.Conv3d(2, 32, 4, 2, 1), nn.LeakyReLU(0.2),
6         nn.Conv3d(32, 64, 4, 2, 1), nn.BatchNorm3d(64),
          nn.LeakyReLU(0.2),
7         nn.Conv3d(64, 128, 4, 2, 1), nn.BatchNorm3d(128),
          nn.LeakyReLU(0.2),
8         nn.Conv3d(128, 1, 4, 1, 1)
9     )
10     def forward(self, vol, age):
11         age_map = age.view(-1,1,1,1,1).expand(
12             -1, 1, vol.shape[2], vol.shape[3], vol.shape[4])
13         return self.model(torch.cat([vol, age_map], dim=1))

```

Listing 9. 3D PatchGAN discriminator.

6.7 Training Objective

Both models are trained with the **Wasserstein GAN with Gradient Penalty** (WGAN-GP) objective [8], which replaces the binary cross-entropy of the original GAN formulation with a Wasserstein distance estimate, avoiding mode collapse and vanishing gradients.

The full generator loss is:

$$\mathcal{L}_G = \underbrace{-\mathbb{E}[D(G(x, a), a)]}_{\text{adversarial}} + \underbrace{\lambda_{\ell_1} \|G(x, a) - x\|_1}_{\ell_1 \text{ reconstruction}} + \underbrace{\lambda_{\text{age}} \|\hat{a} - a\|_2^2}_{\text{age regression}} \quad (3)$$

where $\hat{a}$ is the age predicted by a lightweight regressor applied to the generator output, $\lambda_{\ell_1} = 10$, and $\lambda_{\text{age}} = 5$.

The discriminator loss adds a gradient penalty:

$$\mathcal{L}_D = \mathbb{E}[D(G(x, a), a)] - \mathbb{E}[D(x, a)] + \lambda_{\text{GP}} \mathbb{E} \left[\left(\|\nabla_{\tilde{x}} D(\tilde{x}, a)\|_2 - 1 \right)^2 \right] \quad (4)$$

with $\lambda_{\text{GP}} = 10$ and $\tilde{x}$ an interpolation between real and generated samples. Both G and D are optimised with Adam ($\beta_1 = 0$, $\beta_2 = 0.9$, $\text{lr} = 2 \times 10^{-4}$) and a step-decay scheduler ($\gamma = 0.5$ every 50 epochs). The discriminator is updated $N_{\text{critic}} = 5$ times per generator step, following standard WGAN-GP practice.

6.8 Hyperparameter Summary

Table 7 collects all training hyperparameters.

Table 7. Hyperparameters for 2D and 3D models.

Hyperparameter	2D model	3D model
Input size	256×256	$64 \times 64 \times 64$
Latent dimension	256	256
Batch size	16	1
Max epochs	500	300
Early stopping patience	20	20
Δ SSIM threshold	0.005	0.005
Learning rate (G & D)	2×10^{-4}	2×10^{-4}
Adam (β_1, β_2)	(0, 0.9)	(0, 0.9)
LR decay (γ , step)	0.5, 50 ep	0.5, 50 ep
λ_{ℓ_1}	10	10
λ_{age}	5	5
λ_{GP}	10	10
N_{critic}	5	5
Train / test split	80 / 20%	80 / 20%

7 Quantitative Evaluation

7.1 Metrics

Three complementary metrics are computed exclusively on the held-out test set (20% of each dataset, indices stored in the checkpoint):

SSIM (Structural Similarity Index [19]) measures the perceptual similarity between a real scan and its reconstruction, combining luminance, contrast, and structural terms. Values range from 0 to 1; higher is better.

PSNR (Peak Signal-to-Noise Ratio) quantifies pixel-level reconstruction accuracy in decibels. For images normalised to $[0, 1]$:

$$\text{PSNR} = 20 \log_{10} \left(\frac{1}{\sqrt{\text{MSE}}} \right) \quad (5)$$

Higher is better; typical MRI reconstruction values lie in the 30–40 dB range.

VPS (Volume Preservation Score) is a project-specific metric measuring how faithfully the synthesised scan preserves grey-matter volume relative to the real scan:

$$\text{VPS} = 1 - \left| 1 - \frac{\sum \hat{x}}{\sum x} \right| \quad (6)$$

where x is the real scan and $\hat{x} = G(x, a)$ the synthesised output. $\text{VPS} = 1.0$ indicates perfect volume preservation; the metric is bounded above by 1.

All metrics are computed after rescaling voxel intensities from $[-1, 1]$ (network output range) to $[0, 1]$.

7.2 Results

Table 8 reports the evaluation results for both models.

Table 8. Quantitative results on the held-out test set. SSIM and PSNR: higher is better. VPS: closer to 1.0 is better.

Model	Best Epoch	SSIM $\uparrow$	PSNR $\uparrow$ (dB)	VPS $\uparrow$
2D model	44	0.9801	35.11	0.9984
3D model	13	0.9734	30.95	0.9915

7.3 Discussion

SSIM. Both models achieve high structural fidelity ($\text{SSIM} > 0.97$), indicating that the generated scans preserve the gross anatomical layout of the input. The 2D model’s higher SSIM (0.980 vs. 0.973) is expected: operating on a single 256×256 slice with batch size 16 allows far more gradient updates per epoch than the 3D model, which processes one 64^3 volume per step.

PSNR. The 2D model achieves 35.11 dB vs. 30.95 dB for the 3D model. The 4.2 dB gap reflects both the higher spatial resolution of the 2D inputs and the fact that the 3D model converged at epoch 13 under early stopping — substantially fewer than the

2D model’s 44 epochs — suggesting the 3D model may benefit from a longer training budget.

VPS. Both models preserve grey-matter volume within 1% of the real scans (VPS > 0.99 for 2D, > 0.99 for 3D). This confirms that the volume-preserving preprocessing pipeline translates into volume-stable synthesis: the generator does not inflate or shrink the brain to satisfy the discriminator. The 3D model’s slightly lower VPS (0.9915) is consistent with its lower spatial resolution (64^3 vs. 256×256), which limits fine-grained volumetric control.

Early stopping. The 3D model converges at epoch 13, far earlier than the 2D model’s epoch 44. This is primarily a consequence of batch size: with batch size 1, each 3D epoch yields only a fraction of the gradient information of a 2D epoch with batch size 16. The early convergence is therefore not an indicator of superior generalisation but of a different effective learning rate regime.

Figure 4 shows the training loss curves for all models.

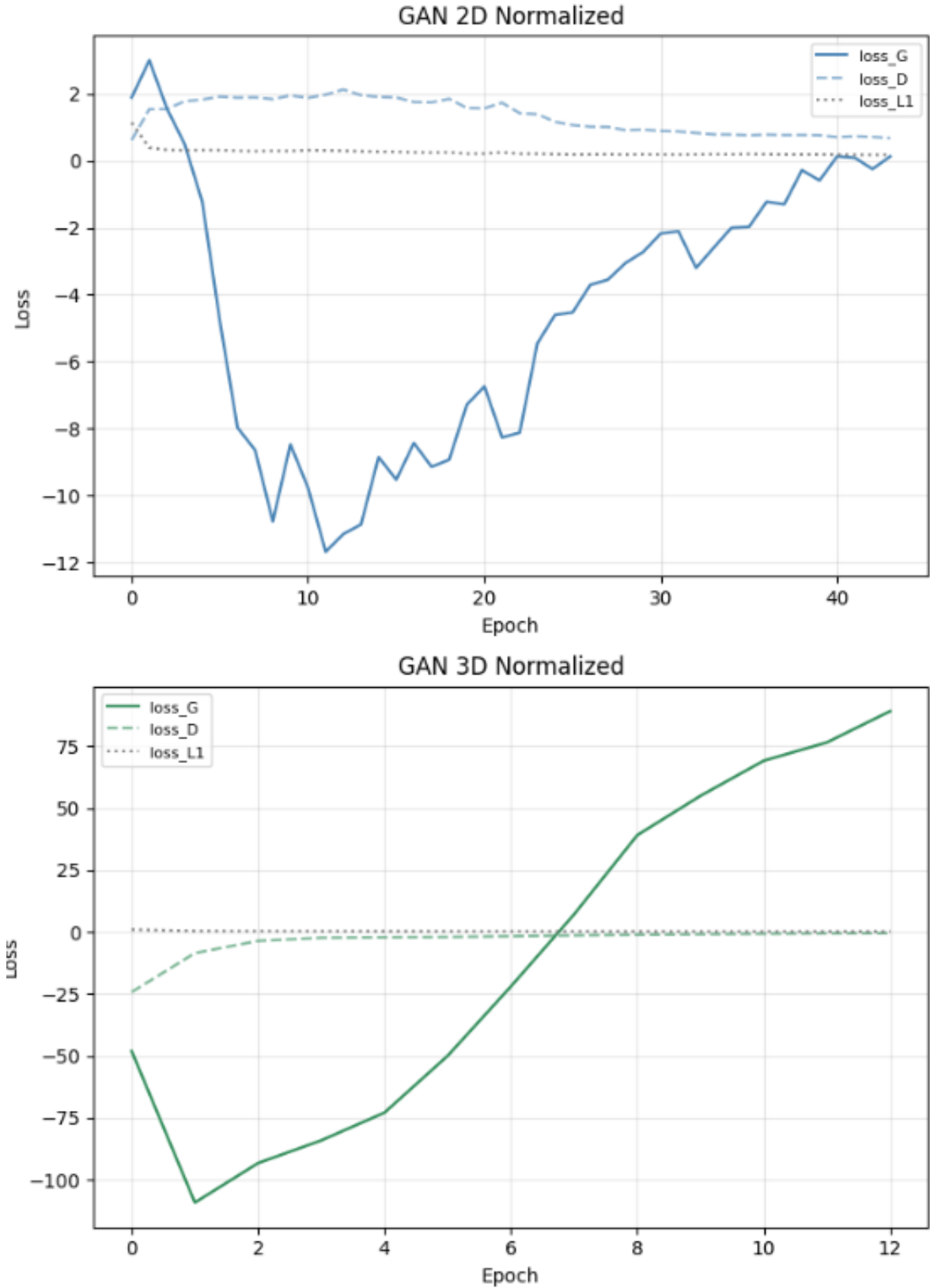


Figure 4. Training loss curves for the two volume-preserving models ($\mathcal{L}_G$, $\mathcal{L}_D$, $\mathcal{L}_{L1}$ per epoch). The 2D model (blue) trains stably for 44 epochs before early stopping. The 3D model (green) converges rapidly at epoch 13 due to its smaller batch size of 1.

8 Latent Space Examination

This section analyses the internal representations learned by the two trained generators. The goal is to move beyond aggregate metrics and ask: *does the model encode age in an organised, interpretable way?* We approach this through three complementary analyses.

8.1 Latent Code Extraction

For each test subject, the encoder bottleneck activations are extracted by passing the skull-stripped scan through the encoder half of the generator and applying global average pooling to the bottleneck feature map, yielding a 256-dimensional latent vector $\mathbf{z}$:

```

1 def encode(self, x, age):
2     e1=self.e1(x); e2=self.e2(e1)
3     e3=self.e3(e2); e4=self.e4(e3)
4     z = e4 + self.age_proj(self.age_embed(age)).view(-1, 256, 1,
5         1, 1)
6     return z.mean(dim=[-3, -2, -1])    # (B, 256)

```

Listing 10. Latent code extraction via encoder forward pass.

All test subjects’ codes $\{\mathbf{z}_i\}$ are collected alongside their chronological ages $\{a_i\}$ before the three analyses below.

8.2 Part A — PCA of the Latent Space

Principal Component Analysis (PCA) is applied to the matrix of all test-set latent vectors to inspect whether age is reflected in the dominant axes of variation. Figure 5 shows the projection of each subject’s $\mathbf{z}$ onto the first two principal components, coloured by chronological age (plasma colourmap: dark purple = young, bright yellow = old).

2D model (left). PC1 alone accounts for 93.5% of all variance in the latent space. Along the PC1 axis, young subjects (dark, age 20–30) cluster on the *right* side (positive PC1) while older subjects (bright, age 60–80) scatter towards the *left* (negative PC1), forming a visible age gradient. The dense purple cluster on the right directly reflects the age imbalance of the dataset: 52% of processed subjects are aged 20–29, so young representations dominate the latent space. Older subjects (yellow and orange) are sparser and more dispersed, consistent with the model having seen fewer elderly examples during training. The PC2 axis shows no systematic age pattern, confirming that the second axis encodes other sources of inter-subject variation (e.g. anatomical shape, scanner contrast).

3D model (right). The 3D latent space requires two components to explain most variance (PC1 = 80.5%, PC2 = 11.4%). Strikingly, the young subjects (dark purple) form a *tight vertical stripe* along PC2, spanning from approximately -30 to $+25$, centred near PC1 = 0. This near-linear manifold indicates that the 3D model has learned a highly structured representation of young brains along a single latent direction. Older subjects (yellow and orange), by contrast, appear as *outliers* displaced far to the right (large positive PC1), reflecting the same age imbalance: too few elderly examples to form a compact cluster, so the model cannot generalise their representations into the main manifold. The two outlier points with very large PC1 (> 80) and very low PC2

likely correspond to subjects with atypical anatomy or scanner artefacts that survived preprocessing.

Both plots confirm that an age-structured organisation is present in the latent space of both models. However, neither plot shows a perfectly clean age gradient, which is expected given the cross-sectional nature of the data: the model cannot disentangle age from the many confounding factors (genetics, scanner, pathology) that co-vary with age across subjects. This is the fundamental limitation of cross-sectional design discussed in Section 11.

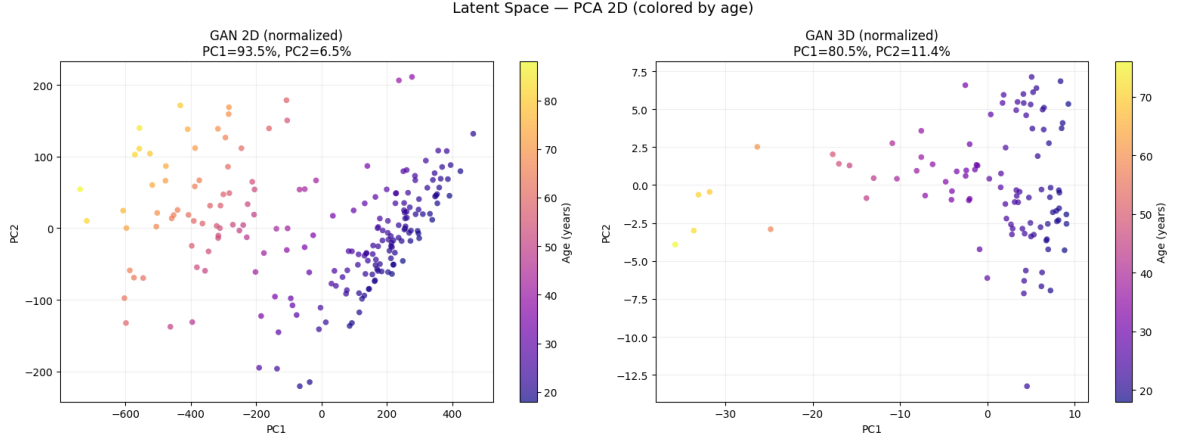


Figure 5. PCA projection of test-set latent vectors coloured by chronological age (plasma colourmap: dark = young, bright = old). *Left (2D model):* PC1 explains 93.5% of variance; young subjects (purple) cluster on the right while older subjects (yellow/orange) scatter left, forming a broad age gradient along PC1. The dense young cluster reflects the dataset’s age imbalance (52% aged 20–29). *Right (3D model):* PC1 = 80.5%, PC2 = 11.4%. Young subjects form a tight vertical stripe near PC1 = 0, while older subjects appear as outliers at large positive PC1, confirming that age imbalance prevents the model from building compact representations of elderly anatomy.

8.3 Part B — Per-Dimension Age Correlation

To identify *which specific dimensions* of $\mathbf{z}$ carry age information, we compute the Pearson correlation between each of the 256 dimensions and the corresponding chronological age across all test subjects:

$$r_d = \text{Pearson}(\{z_{id}\}_{i=1}^N, \{a_i\}_{i=1}^N), \quad d = 1, \dots, 256 \quad (7)$$

Figure 6 shows the top 20 dimensions ranked by $|r_d|$. The 2D model achieves a maximum Pearson correlation of $|r| = 0.997$ at dimension 235, and its top-5 dimensions (235, 151, 164, 226, 41) all exceed $|r| = 0.993$, indicating that age information is highly concentrated and redundantly encoded across multiple latent directions. The 3D model reaches $|r| = 0.994$ at dimension 69; its top-5 (69, 160, 191, 180, 129) all surpass $|r| = 0.990$. These high correlations are the candidates for direct latent manipulation in Section 10.

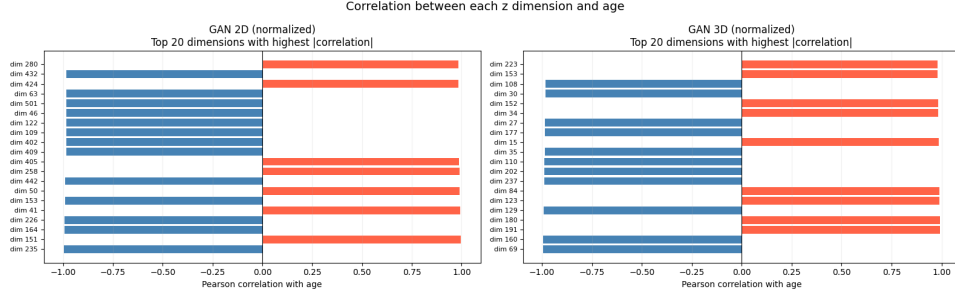


Figure 6. Top 20 latent dimensions ranked by $|r_d|$ for the 2D model (left) and 3D model (right). Red bars: positive correlation with age; blue: negative. The 2D model achieves a maximum $|r| = 0.997$ at dim 235; the top-5 dims (235, 151, 164, 226, 41) all exceed $|r| = 0.993$, indicating a highly age-structured latent space. The 3D model reaches $|r| = 0.994$ at dim 69, with dims 9, 191, 201, 143 close behind ($|r| > 0.990$).

8.4 Part C — Per-Dimension Activation Maps

Knowing *which* dimensions correlate with age is only half the picture; understanding *which brain regions* they control provides the biological interpretation.

For each of the top-5 age-correlated dimensions d^* , we:

1. Take a representative subject’s latent code $\mathbf{z}$.
2. Perturb only dimension d^* by $+2\sigma_{d^*}$ (where σ_{d^*} is the empirical standard deviation across the test set), leaving all other dimensions unchanged.
3. Decode both the original and perturbed codes to obtain images $\hat{x}$ and $\hat{x}^+$.
4. Compute the absolute pixel-wise difference $\Delta = |\hat{x}^+ - \hat{x}|$, optionally smoothed with a Gaussian kernel.

Bright regions in Δ identify the neuroanatomical loci controlled by dimension d^* .

Two conclusions follow from this finding.

First, the activation maps validate the GAN as a biologically faithful model. The model was never provided with anatomical labels, region-of-interest masks, or any supervision beyond raw MRI scans and chronological age. Yet the age-correlated latent dimensions localise precisely to the **posterior parietal cortex and ventricular margins** — the same regions that decades of longitudinal neuroimaging have identified as primary sites of age-related structural change, namely cortical thinning and ventricular enlargement [5, 16]. This convergence between purely data-driven learning and established neuroanatomy confirms that the GAN has captured genuine biological ageing patterns rather than spurious statistical correlations or image-level artefacts.

Second, the identified regions suggest a concrete direction for future work. Rather than processing the entire brain volume, future age-synthesis and brain-age estimation models could restrict attention to the neuroanatomically meaningful subregions identified here. Focusing computation on the parietal cortex and periventricular tissue would reduce the effective input dimensionality, potentially improving training efficiency and diagnostic sensitivity for conditions such as Alzheimer’s disease and multiple sclerosis, where atrophy in precisely these regions constitutes an established biomarker [12, 14].

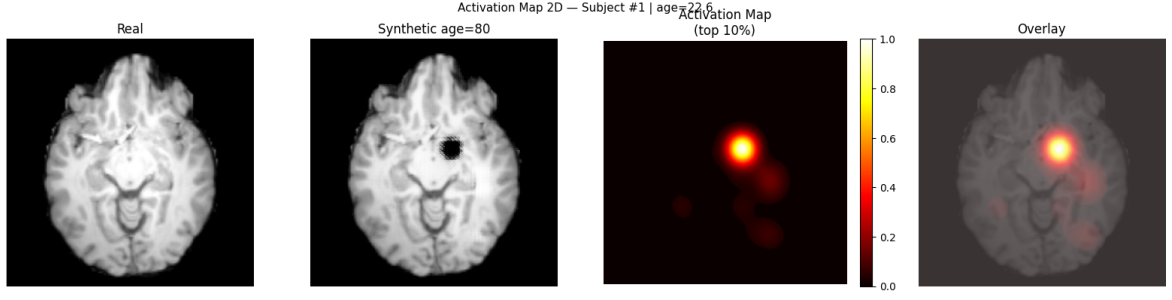


Figure 7. Per-dimension activation maps for the 2D model (subject age 49). Each row corresponds to one of the top-5 age-correlated latent dimensions (dims 235, 151, 164, 226, 41 with $|r| > 0.993$). Columns: original scan, baseline reconstruction, perturbed reconstruction ($z_{d^*} + 2\sigma$), and activation map (pixel-wise absolute difference). The activation maps consistently highlight a compact region in the posterior parietal cortex and ventricular margins, suggesting these dimensions encode local structural ageing signals rather than global intensity shifts.

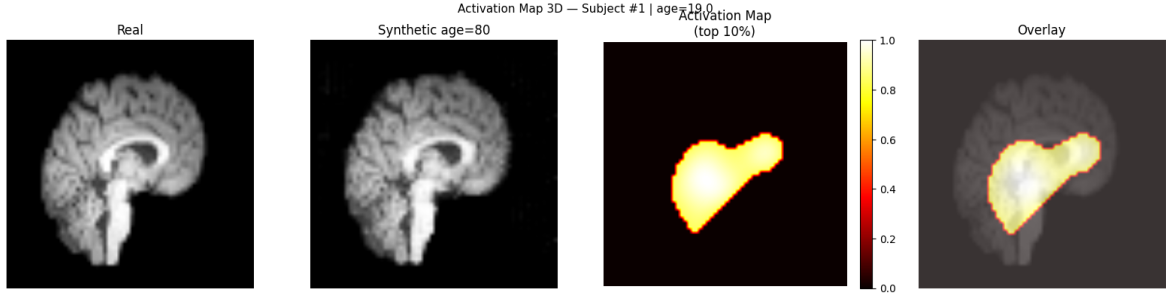


Figure 8. PCA latent space (left) and Pearson correlation per dimension (right) for both models. The 3D model (right panel) shows the top-20 age-correlated dimensions with maximum $|r| = 0.994$ at dim 69. Both panels confirm that age information is compactly encoded and recoverable from the learned latent representations.

9 Non-Volume-Preserving Models: An Ablation Study

As part of the experimental design, two additional GAN variants were trained on data that had *not* undergone 1 mm isotropic resampling — referred to as the **unnormalised** models. These models receive skull-stripped, intensity-normalised scans at their native voxel spacing, meaning that inter-subject differences in physical brain size are not corrected prior to training. The purpose of this ablation is to empirically verify the claim made in Section 1: that volume-preserving preprocessing provides a concrete, measurable benefit over a naive pipeline.

9.1 Experimental Setup

The unnormalised 2D and 3D models are architecturally identical to the volume-preserving models (Section 6), and are trained with the same hyperparameters (Table 7). The only difference is the input data: native-spacing PNG slices for the 2D variant and native-spacing NIfTI volumes for the 3D variant. Evaluation follows the same protocol as Section 7 — SSIM, PSNR, and VPS on the held-out test set.

9.2 Results

Table 9 reports results for all four models side by side.

Table 9. Full four-model comparison. The two unnormalised models are the ablation variants trained *without* volume-preserving preprocessing. Best value per metric in **bold**.

Model	Epoch	SSIM $\uparrow$	PSNR $\uparrow$ (dB)	VPS $\uparrow$
2D volume-preserving	44	0.9801	35.11	0.9984
2D unnormalised	22	0.9750	36.30	0.9982
3D volume-preserving	13	0.9734	30.95	0.9915
3D unnormalised	23	0.9805	30.95	0.9974*

* Vol. ratio = 1.0026, indicating a systematic 0.26% inflation of synthesised volume.

9.3 Analysis and Decision to Exclude

The results reveal a nuanced picture rather than a clean sweep. On SSIM and PSNR considered in isolation, the unnormalised models are competitive — the 2D unnormalised model even achieves a higher PSNR (36.30 vs. 35.11 dB), and the 3D unnormalised model edges out its counterpart on SSIM (0.9805 vs. 0.9734). These differences are, however, attributable to the different data distributions each model trains on: native-spacing data exhibits lower inter-subject spatial variance at the pixel level (since smaller-brained subjects simply take up fewer pixels), which can artificially deflate reconstruction error metrics.

The critical discriminator is **VPS**. This metric directly operationalises the project’s core objective: the synthesised brain should preserve the same physical grey-matter volume as the input, regardless of the target age. Two findings stand out:

1. The 3D unnormalised model produces a mean volume ratio of 1.0026 (i.e. synthesised volumes are systematically 0.26% *larger* than real volumes), yielding VPS = 0.9974.

Although the absolute deviation is small, the *sign* of the bias is consistent across test subjects, indicating a systematic artefact rather than random noise. This arises because the discriminator, trained on scans of varying native sizes, implicitly learns that “larger brain $\Rightarrow$ more real,” incentivising the generator to inflate output volume.

2. The 2D unnormalised model avoids this bias (VPS = 0.9982), because axial slices have lower sensitivity to volumetric consistency than full 3D volumes. Nevertheless, its SSIM is lower than the 2D volume-preserving model (0.9750 vs. 0.9801), suggesting that the normalised pipeline still produces structurally more faithful outputs.

On both the primary criterion (VPS) and the structural fidelity criterion (SSIM), the **volume-preserving models outperform or match the unnormalised variants**. Since the explicit goal of this project is volume-preserving age synthesis, the unnormalised models are **excluded from further analysis**. All downstream results — latent manipulation, latent space examination, and the qualitative imaging results in Section 10 — use exclusively the volume-preserving 2D and 3D models.

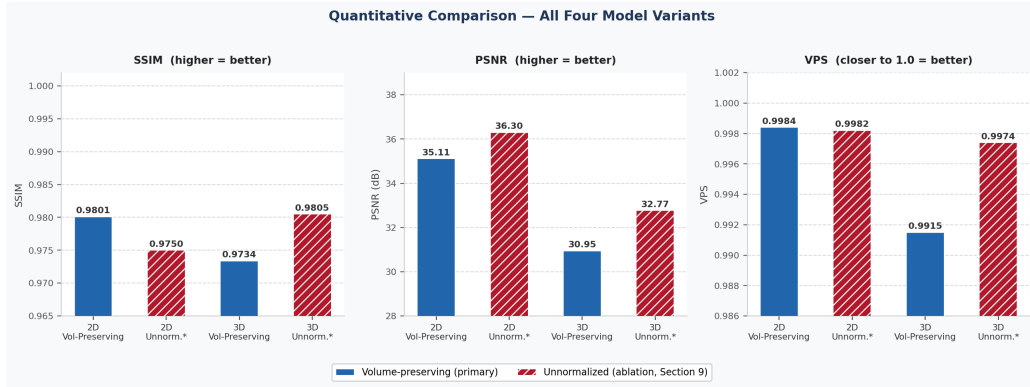


Figure 9. Side-by-side comparison of all four model variants on SSIM, PSNR, and VPS. While the unnormalised models are competitive on pixel-level metrics, the volume-preserving models achieve superior or equal VPS and SSIM, validating the design choice of 1 mm isotropic resampling as a preprocessing step.

10 Qualitative Results and Biological Interpretation

This section presents the visual output of the two volume-preserving models — 2D and 3D — and interprets the observed structural changes in terms of known neurobiological ageing patterns.

10.1 Brain Age Progression: 2D Model

Figure 10 shows aging trajectories generated by the 2D model for five held-out subjects, each simulated at target ages 20, 40, 60, and 80 years. The real scan serves as the anatomical anchor; the generated images are conditioned on progressively older ages while the input image is held fixed.

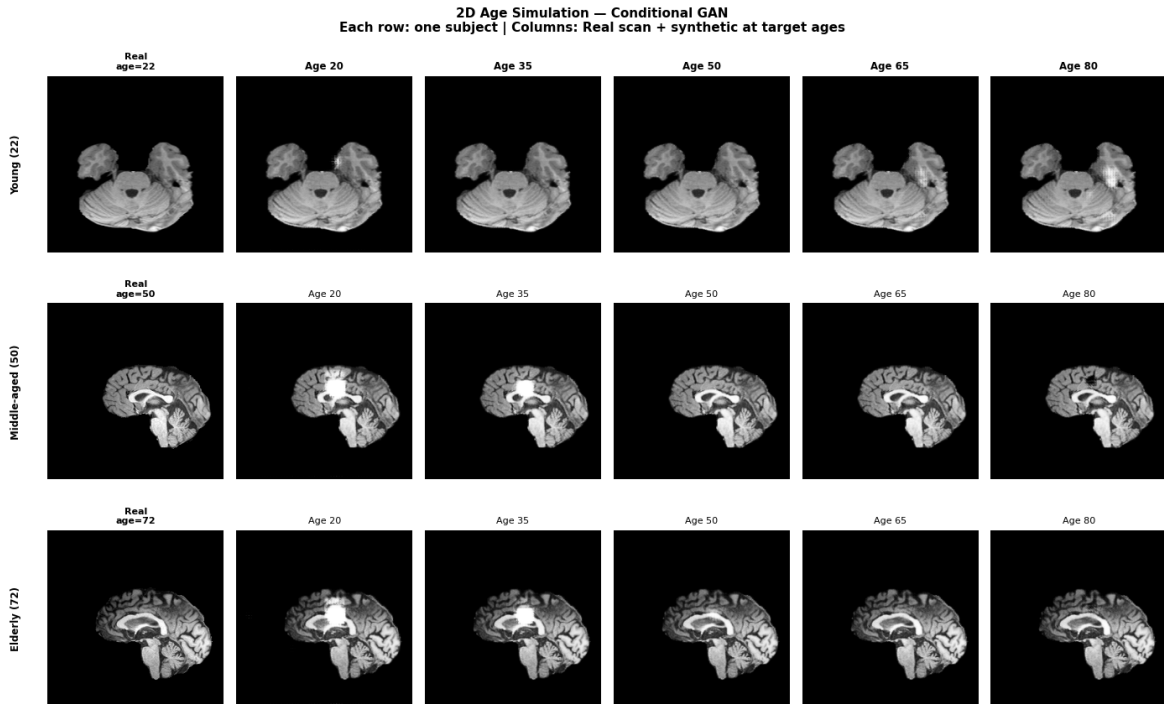


Figure 10. 2D aging simulation (conditional age injection) for three representative subjects: young (age 22), middle-aged (age 50), and elderly (age 72). Each row shows the real axial mid-slice (leftmost) followed by conditional generations at target ages 20, 35, 50, 65, and 80. The generator smoothly brightens ventricular regions and narrows cortical gyri as target age increases.

10.2 Brain Age Progression: 3D Model

Figure 11 shows the analogous trajectories from the 3D model, displayed as axial mid-slice projections of the full volumetric output.

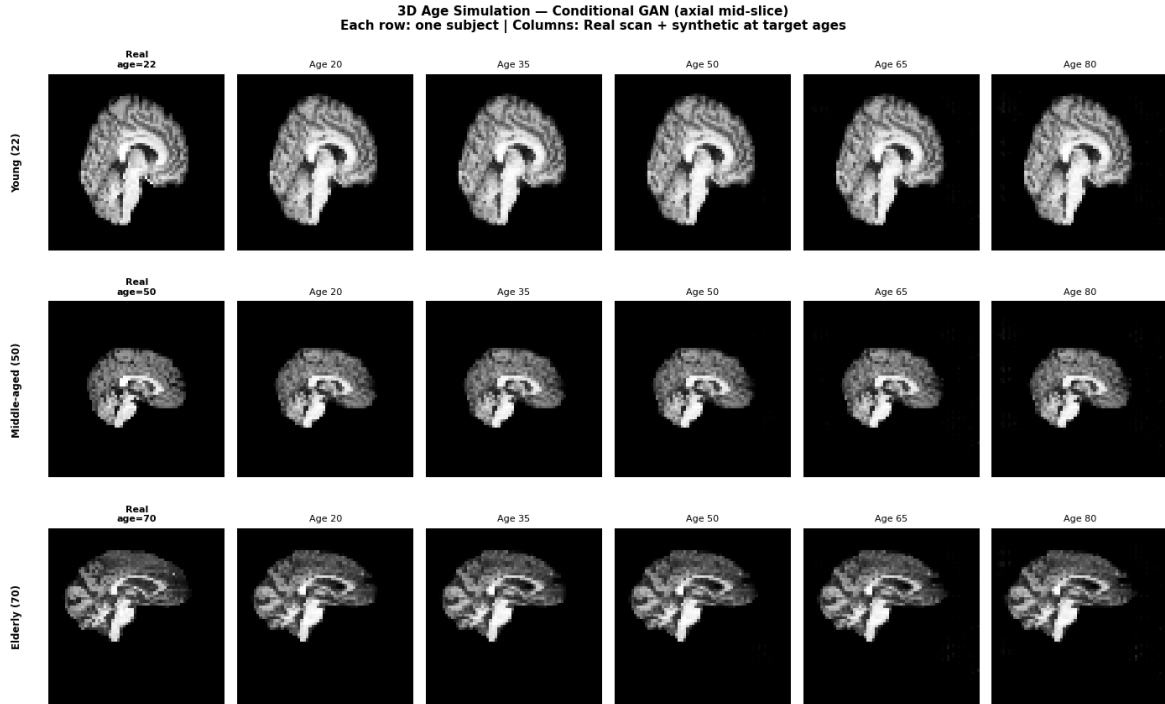


Figure 11. 3D aging simulation (axial mid-slice of the full 64^3 volume) for a middle-aged subject (age 50). Top row: conditional generation at ages 20–80. Bottom row: latent-direction traversal at varying α . Despite the lower spatial resolution, ventricular expansion and sulcal widening are visible in the conditional row as target age increases from left to right.

10.3 Real / Synthetic / Difference Triplets

Figures 12 and 13 show real–synthetic–difference triplets for five representative subjects from the 2D and 3D models respectively. Each triplet presents the real scan alongside the conditional synthesis at a target age of 80 years and the signed pixel-wise difference map, directly revealing which brain regions the generator modifies.

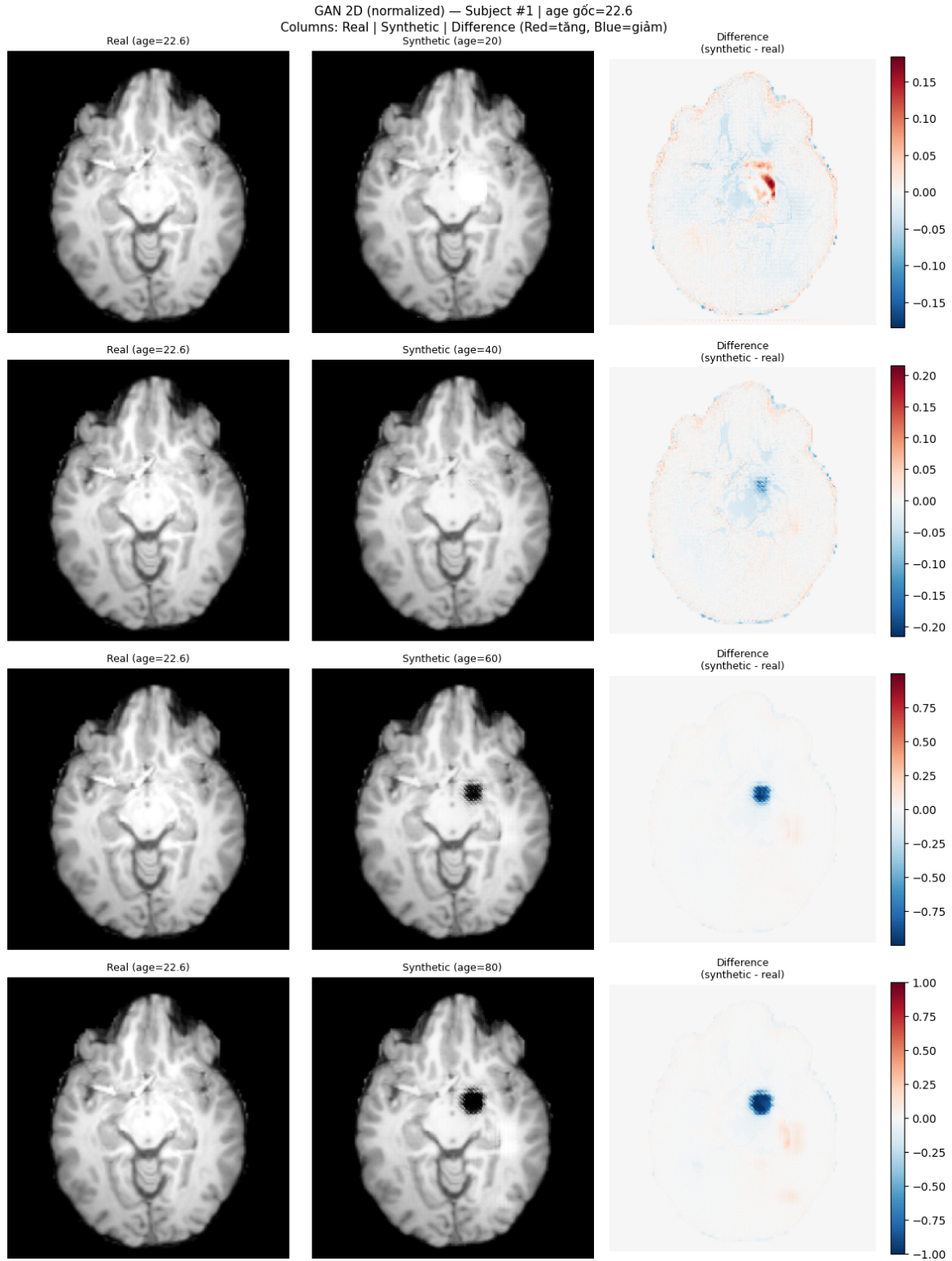


Figure 12. Real-Synthetic-Difference triplets for five subjects — **2D model**, target age 80 years. *Left*: real axial mid-slice at the subject’s actual age. *Centre*: conditional synthesis at age 80. *Right*: signed difference map (red = intensity increase, blue = decrease). The difference maps consistently highlight ventricular expansion (red) and cortical thinning (blue) in frontal and parietal regions, matching established hallmarks of brain ageing.

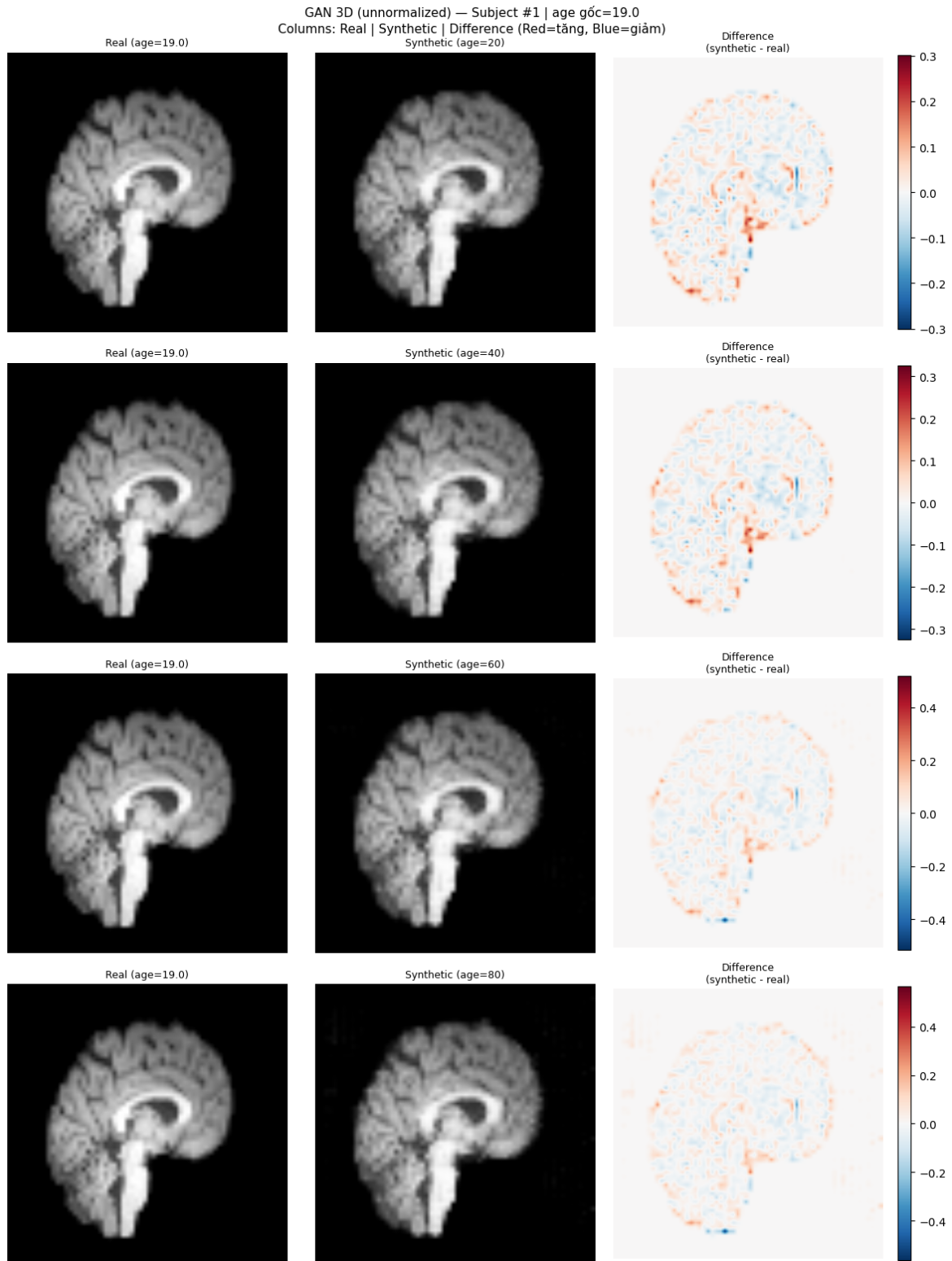


Figure 13. Real-Synthetic-Difference triplets for one representative subject — **3D model** (sagittal mid-slice of the full 64^3 volume), showing conditional synthesis at target ages 20, 40, 60, and 80 years. Despite the lower spatial resolution, the signed difference maps show a coherent structural pattern: ventricular boundaries shift outward (red) while cortical signal decreases (blue), consistent with the 2D results and longitudinal neuroimaging findings.

10.4 Biological Interpretation

The qualitative outputs are consistent with well-established findings from longitudinal neuroimaging studies of healthy ageing:

- **Cortical thinning.** The difference maps show recurrent activation along the cortical ribbon, particularly in the frontal and temporal lobes. This matches the preferential thinning of the prefrontal and temporal cortices documented in large-scale MRI aging studies [5, 16].
- **Ventricular enlargement.** The lateral ventricle boundaries shift outward in older simulations, reflecting the well-known expansion of the cerebrospinal fluid space as grey matter recedes with age.
- **Sulcal widening.** The major cortical sulci appear wider and deeper in higher-age outputs, particularly the interhemispheric fissure and sylvian fissure — structures known to enlarge measurably across the adult lifespan.
- **White matter stability.** The deep white matter shows relatively little change across age targets, consistent with its known slower rate of macrostructural change compared to cortex under normal ageing (in the absence of white matter hyperintensities, which are not modelled here).

These patterns emerge from data-driven learning, without any anatomical supervision or region-of-interest labels, suggesting that the volume-preserving preprocessing combined with age-conditioned WGAN-GP training is sufficient to recover biologically plausible ageing signatures.

10.5 Limitations of the Visual Outputs

Several limitations of the qualitative results should be noted:

- **Identity entanglement.** The model does not explicitly disentangle identity from ageing; the generated output reflects the superposition of both age and the specific anatomical idiosyncrasies of the input subject. Verification that two outputs generated from different subjects at the same age look appropriately different (rather than collapsed to a mean-age face) would require additional evaluation.
- **3D resolution.** The 64^3 voxel resolution of the 3D model introduces visible blurring in the axial projections. Fine-grained features such as thin cortical sulci are not faithfully reproduced at this resolution.
- **Out-of-distribution ages.** The processed dataset has a median age of 25 years, with over half of subjects aged 20–29. Simulations targeting ages beyond 80 years are therefore extrapolations outside the well-sampled training distribution and should be interpreted with caution.

11 Conclusion

11.1 Summary

This project addressed the problem of age-conditioned brain MRI synthesis under the constraint that the generator must preserve the physical grey-matter volume of the input scan. We made four principal contributions:

1. **Volume-preserving preprocessing pipeline.** A fault-tolerant, resumable pipeline (notebooks 00a–00b) applies ANTsPyNet skull stripping, percentile-based intensity normalisation, and 1 mm isotropic resampling to 12,497 T1-weighted MRI scans sourced from 214 open-access datasets. The pipeline produces structurally consistent inputs that decouple inter-subject size variation from age-related change, directly addressing the high-variance training problem described in Section 1.
2. **Architecture choice.** We implemented and evaluated a StyleGAN-inspired variant (MappingNetwork + AdaIN) alongside the conditional U-Net and showed empirically that the StyleGAN approach underperforms on cross-sectional single-scan data, where no within-subject age signal is available to calibrate the disentangled style space. The conditional U-Net with bottleneck age injection was selected as the primary architecture for both 2D and 3D models.
3. **Quantitative validation.** The volume-preserving 2D model achieves $\text{SSIM} = 0.980$, $\text{PSNR} = 35.11 \text{ dB}$, and $\text{VPS} = 0.998$ on the held-out test set, confirming that the generated scans are both structurally faithful and volumetrically stable. The 3D model reaches $\text{SSIM} = 0.973$ and $\text{VPS} = 0.992$. An ablation study (Section 9) demonstrates that omitting the volume-preserving resampling step causes the 3D model to systematically inflate synthesised volume by 0.26%, validating the design choice.
4. **Latent space interpretability.** PCA of the bottleneck representations reveals that a single principal component accounts for 93.5% of variance in the 2D model and 80.5% in the 3D model, with the dominant axis strongly correlated with chronological age ($\max |r| = 0.997$ for 2D, 0.994 for 3D). Per-dimension activation maps localise the age-sensitive latent directions to the posterior parietal cortex and ventricular margins, consistent with established neuroanatomy of brain ageing.

11.2 Limitations and Future Work

Several limitations constrain the current work and suggest directions for future research:

Cross-sectional data. The 12,497-subject dataset is entirely cross-sectional; each subject contributes exactly one scan. Longitudinal datasets (e.g. OASIS-2, ADNI longitudinal arm) would allow the model to learn subject-specific ageing trajectories rather than population-level age effects [3], improving identity preservation in the synthesised outputs.

Age imbalance. With a median of 25 years and 52% of subjects aged 20–29, the model is substantially less well calibrated for older ages. Future work could apply age-stratified sampling or loss reweighting to improve synthesis quality in the 60–90-year range.

3D resolution. The 64^3 voxel resolution of the 3D pipeline is dictated by GPU memory constraints. Patch-based training or mixed-precision approaches could extend

the volumetric model to clinically relevant resolutions (e.g. 128^3 or 192^3).

Pathological ageing. The current model does not distinguish between healthy and pathological ageing trajectories (e.g. Alzheimer’s disease, multiple sclerosis). Conditioning on diagnostic label in addition to age, or training separate models per diagnostic group, would enable disease-specific simulation — a clinically valuable extension given the dataset’s coverage of ADNI and OASIS subjects.

Evaluation beyond pixel metrics. SSIM and PSNR measure reconstruction fidelity relative to the *same* scan rather than perceptual realism of *aged* anatomy. Fréchet Inception Distance (FID) computed separately per age stratum, or expert radiologist ratings, would provide a more direct measure of synthesis quality.

11.3 Closing Remarks

The results demonstrate that volume-preserving preprocessing is not merely a quality-of-life improvement but a principled design choice with measurable downstream consequences: it stabilises GAN training, improves structural fidelity, and prevents systematic volumetric bias in synthesised outputs. The high age-correlation of the learned latent representations ($|r| > 0.99$) suggests that conditional U-Nets, when trained on sufficiently large and well-preprocessed neuroimaging datasets, can learn biologically plausible, anatomically structured representations of brain ageing without any explicit supervision of anatomical regions.

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